Name: ..... Due: **Next Class / Friday**

C3 Early Multitrack Recordings HW 1 – The Beatles 'Eleanor Rigby' & 4-Track Tape

Artist: The Beatles (Producer: George Martin, Engineer: Geoff Emerick)
https://www.youtube.com/watch?v=HuS5NuXRb5Y

1. In 1966, The Beatles recorded "Eleanor Rigby" onto a Studer J37 4-track tape machine at Abbey Road. Identify the track allocation for each instrument or vocal in the table below. (3)

Track Element	Assigned Tape Track & Panning
String Octet (4 Violins, 2 Violas, 2 Cellos)	Track 1 & 2 (Panned Left & Right)
Paul McCartney Lead Vocal	Track 3 (Centre / Hard Right in Stereo LP)
John Lennon & George Harrison Backing Vocals	Track 4 (Centre / Hard Left)
Vocal Double / Ad-Lib Strings	

2. What rhythmic string articulation can be heard driving the rhythm from 0'04"? (1)

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3. What year was this track recorded in, and state ONE studio production technique that proves it was recorded on 4-track tape rather than direct-to-mono? (2)

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4. Abbey Road technical engineers utilized the Studer J37 with Sel-Sync. Explain why Sel-Sync was essential during overdubbing and what would happen if the artist monitored off the playback head. (4)

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5. When recording ambitious multitrack arrangements with only 4 tracks, engineers performed "reduction mixing" (bouncing down). Explain how a 4-to-1 reduction mix works and state ONE sonic trade-off of this technique. (4)

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6. In "Eleanor Rigby", Geoff Emerick famously placed Neumann microphones closer to the strings than classical EMI protocol permitted. Describe how you would close-mic an acoustic cello or upright bass in a modern studio: (4)

Choice of Microphone & Polar Pattern (2)
Microphone: Polar Pattern:
Positioning and Distance of Microphone (2)

Total:	**/18**